

Assignment 2: Coding Basics

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OVERVIEW

This exercise accompanies the lessons/labs in Environmental Data Analytics on coding basics.

Directions

1. Rename this file `<FirstLast>_A02_CodingBasics.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.
6. After Knitting, submit the completed exercise (PDF file) to Canvas.

Basics, Part 1

1. Generate a sequence of numbers from one to 55, increasing by fives. Assign this sequence a name.
2. Compute the mean and median of this sequence.
3. Ask R to determine whether the mean is greater than the median.
4. Insert comments in your code to describe what you are doing.

```
#1.  
seq = seq(1, 55, 5) #generating the sequence called "seq" from `1` to `55` with an increment of `5`.  
seq #display "seq"
```

```
## [1] 1 6 11 16 21 26 31 36 41 46 51
```

```
#2.  
mean(seq) # show the mean of "seq"
```

```
## [1] 26
```

```
median(seq) # show the median of "seq"
```

```
## [1] 26
```

```
#3.  
mean(seq) > median(seq) # ask r if mean is larger than median
```

```
## [1] FALSE
```

Basics, Part 2

5. Create three vectors, each with four components, consisting of (a) student names, (b) test scores, and (c) whether they are on scholarship or not (TRUE or FALSE).
6. Label each vector with a comment on what type of vector it is.
7. Combine each of the vectors into a data frame. Assign the data frame an informative name.
8. Label the columns of your data frame with informative titles.

```
a = c("Alex", "Betty", "Catherine", "David") # character  
b = c(92,95,89,87) # numeric  
c = c(TRUE,TRUE,FALSE,FALSE) # logical  
  
dataframe = data.frame(a,b,c)  
names(dataframe) = c("name", "test_score", "scholarship")
```

9. QUESTION: How is this data frame different from a matrix?

Answer: A matrix must contain elements of the same data type. A data frame can have columns with different data types.

10. Create a function with one input. In this function, use `if...else` to evaluate the value of the input: if it is greater than 50, print the word “Pass”; otherwise print the word “Fail”.
11. Create a second function that does the exact same thing as the previous one but uses `ifelse()` instead of `if...else`.
12. Run both functions using the value 52.5 as the input
13. Run both functions using the **vector** of student test scores you created as the input. (Only one will work properly...)

```
#10. Create a function using if...else  
grader10 = function(grade){  
  if (grade > 50){  
    return("Pass")  
  }  
  else {  
    return("Fail")  
  }  
}  
  
#11. Create a function using ifelse()  
grader11 = function(grade){  
  ifelse (grade > 50, "Pass", "Fail")  
}  
  
#12a. Run the first function with the value 52.5  
grader10(52.5)
```

```
## [1] "Pass"
```

```
#12b. Run the second function with the value 52.5  
grader11(52.5)
```

```
## [1] "Pass"
```

```
#13a. Run the first function with the vector of test scores  
#grader10(b)  
#13b. Run the second function with the vector of test scores  
grader11(b)
```

```
## [1] "Pass" "Pass" "Pass" "Pass"
```

14. QUESTION: Which option of `if...else` vs. `ifelse` worked? Why? (Hint: search the web for “R vectorization”)

Answer: The `if/else` control statement is not vectorized. It is designed to evaluate only a single condition and make one decision based on that. The `ifelse()` function, on the other hand, is vectorized. It is specifically designed to perform an element-by-element test. It takes a vector of conditions and returns a vector of results of the same length.

NOTE Before knitting, you’ll need to comment out the call to the function in Q13 that does not work. (A document can’t knit if the code it contains causes an error!)